

practical session

1.)

```
pooh — nano classproblem1.c — 120x30
UW PICO 5.09 File: classp

#include <stdio.h>

int main()
{
    int x = 10;
    printf("Value of x = %d\n", x);
    printf("Virtual address of x = %p\n", (void *)&x);

    return 0;
}
```

```
[pooh@Kandulas-MacBook-Air ~ % nano classproblem1.c
[pooh@Kandulas-MacBook-Air ~ % gcc classproblem1.c -o classproblem1
[pooh@Kandulas-MacBook-Air ~ % ./classproblem1
Value of x = 10
Virtual address of x = 0x16d1b32c8
```

2.)

```
pooh — nano stack.c — 120x30
UW PICO 5.09 File: stack.c

#include <stdio.h>
#include <stdlib.h>

int global = 10;
int bss_variable;

int main() {
    int local = 20;

    int *heap = malloc(sizeof(int));
    *heap = 30;

    printf("Code address      : %p\n", (void *)main);
    printf("Data address       : %p\n", (void *)&global);
    printf("BSS address        : %p\n", (void *)&bss_variable);
    printf("Heap address       : %p\n", (void *)&heap);
    printf("Stack address      : %p\n", (void *)&local);

    free(heap);

    return 0;
}
```

```
[pooh@Kandulas-MacBook-Air ~ % nano stack.c
[pooh@Kandulas-MacBook-Air ~ % gcc stack -o stack
clang: error: no such file or directory: 'stack'
clang: error: no input files
[pooh@Kandulas-MacBook-Air ~ % gcc stack.c -o stack
[pooh@Kandulas-MacBook-Air ~ % ./stack
Code address      : 0x104340548
Data address      : 0x104348000
BSS address       : 0x104348004
Heap address      : 0x1049d17f0
Stack address     : 0x16babf2e8
```

3.)

```
#include <stdio.h>

int main() {

    int page_table[4] = {1, -1, 3, -1};

    int page;

    printf("Enter page number (0-3): ");
    scanf("%d", &page);

    if (page_table[page] == -1) {
        printf("Page Fault!\n");
        printf("Page must be loaded into memory.\n");
    }
    else {
        printf("Page is already in memory.\n");
        printf("Physical frame = %d\n", page_table[page]);
    }

    return 0;
}
```

```
pooh@Kandulas-MacBook-Air ~ % nano pagetable_pagefaultsimulation.c
pooh@Kandulas-MacBook-Air ~ % gcc pagetable_pagefaultsimulation.c -o pagetable_pagefaultsimulation
pooh@Kandulas-MacBook-Air ~ % ./pagetable_pagefaultsimulation
zsh: no such file or directory: ./pagetable_pagefaultsimulation
pooh@Kandulas-MacBook-Air ~ % ./pagetable_pagefaultsimulation
Enter page number (0-3): 2
Page is already in memory.
Physical frame = 3
```

4.)

```
UW PICO 5.09 File: vatopa.c

#include <stdio.h>

int main() {

    int page_table[] = {3, 5, 1, 7};
    int page_size = 100;

    int virtual_address = 250;

    int page = virtual_address / page_size;
    int offset = virtual_address % page_size;

    int frame = page_table[page];

    int physical_address = frame * page_size + offset;

    printf("Virtual Address = %d\n", virtual_address);
    printf("Page Number      = %d\n", page);
    printf("Offset           = %d\n", offset);
    printf("Frame Number      = %d\n", frame);
    printf("Physical Address = %d\n", physical_address);

    return 0;
}
```

```

[pooh@Kandulas-MacBook-Air ~ % nano vatopa.c
[pooh@Kandulas-MacBook-Air ~ % gcc vatopa.c -o vatopa
[pooh@Kandulas-MacBook-Air ~ % ./vatopa
Virtual Address = 250
Page Number      = 2
Offset           = 50
Frame Number     = 1
Physical Address = 150
pooh@Kandulas-MacBook-Air ~ % █

```

5.)

```

#include <stdio.h>
#include <stdlib.h>

int main() {

    int *p;

    p = malloc(5 * sizeof(int));

    if (p == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    for (int i = 0; i < 5; i++) {
        p[i] = (i + 1) * 10;
    }

    for (int i = 0; i < 5; i++) {
        printf("%d ", p[i]);
    }

    printf("\n");

    free(p);

    return 0;
}
█

```

```

Last login: Thu Sep 17 09:10:59 on ttys000
[pooh@Kandulas-MacBook-Air ~ % nano malloc.c
[pooh@Kandulas-MacBook-Air ~ % gcc malloc.c -o malloc
[pooh@Kandulas-MacBook-Air ~ % ./malloc
10 20 30 40 50
pooh@Kandulas-MacBook-Air ~ % █

```

6.)

```
C
#include <stdio.h>

int main() {

    int *p = NULL;

    printf("Before accessing memory\n");

    *p = 100;

    printf("After accessing memory\n");

    return 0;
}
```

```
[pooh@Kandulas-MacBook-Air ~ % nano nullpointer.c
[pooh@Kandulas-MacBook-Air ~ % gcc nullpointer.c -o nullpointer
[pooh@Kandulas-MacBook-Air ~ % ./nullpointer
Before accessing memory
zsh: segmentation fault ./nullpointer
pooh@Kandulas-MacBook-Air ~ %
```